

Two 60° angles, an isosceles cevian, and a hidden circle

the invariant $AD - BD = DE$, then the power of a point

Problem. In triangle ABC , the point D lies on side BC and the point E lies on side AC . Suppose

$$AD = DC, \quad BD = 8, \quad DE = 7, \quad \angle BAC = \angle ADE = 60^\circ.$$

Find AB .

$$AB = \frac{161}{13}.$$

Overview. The solution has two independent halves, and each one produces an integer.

The first half determines AD . Setting $\alpha = \angle DAC$, the isosceles condition $AD = DC$ makes $\angle ADB = 2\alpha$, and the two triangles ABD and AED turn out to share the angle $120^\circ - \alpha$ opposite to AD . The sine rule in each then gives AD and BD as multiples of DE , and the difference collapses by a sum-to-product identity:

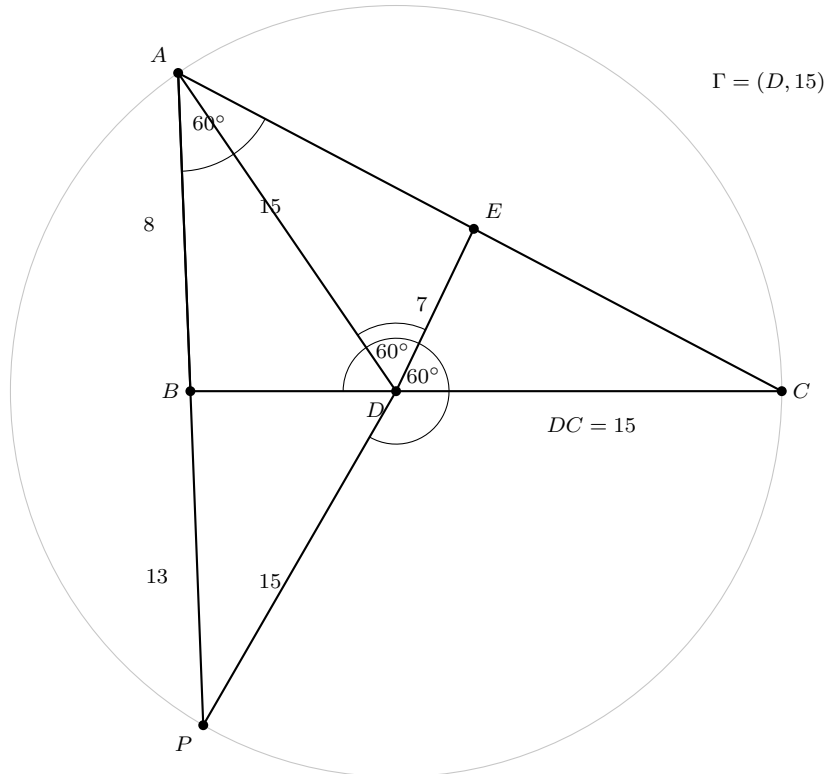
$$AD - BD = DE \cdot \frac{\sin(120^\circ - \alpha) - \sin(60^\circ - \alpha)}{\sin \alpha} = DE.$$

So $AD = BD + DE = 8 + 7 = 15$, independently of α .

The second half uses the circle Γ centred at D with radius $AD = 15$. Both A and C lie on it. Let P be the second intersection of line AB with Γ . The inscribed angle $\angle PAC = 60^\circ$ gives the central angle $\angle PDC = 120^\circ$, hence $\angle PDB = 60^\circ$; the law of cosines in triangle PDB returns $PB = 13$; and the power of the point B with respect to Γ gives

$$AB \cdot PB = (AD - BD)(AD + BD) = 7 \cdot 23 = 161.$$

1. The configuration



Write $\alpha = \angle DAC$. The data are

$$AD = DC, \quad BD = 8, \quad DE = 7, \quad \angle BAC = \angle ADE = 60^\circ,$$

with B, D, C collinear in that order and E on segment AC .

Angle chase.

- Triangle ADC is isosceles with $AD = DC$, so $\angle ACD = \angle DAC = \alpha$.
- $\angle ADB$ is the exterior angle of triangle ADC at D , hence

$$\angle ADB = \angle DAC + \angle ACD = 2\alpha. \quad (1)$$

- Since E lies on AC , the ray AE is the ray AC , so $\angle DAE = \angle DAC = \alpha$, and $\angle BAD = 60^\circ - \alpha$.
- In triangle ADE , with $\angle ADE = 60^\circ$,

$$\angle AED = 180^\circ - 60^\circ - \alpha = 120^\circ - \alpha. \quad (2)$$

- In triangle ABD , with (1),

$$\angle ABD = 180^\circ - \angle ADB - \angle BAD = 180^\circ - 2\alpha - (60^\circ - \alpha) = 120^\circ - \alpha. \quad (3)$$

Comparing (2) and (3):

$$\boxed{\angle ABD = \angle AED = 120^\circ - \alpha.} \quad (4)$$

In both triangles ABD and AED the side AD is opposite an angle of the same size, so the two sine rules share a common ratio.

2. The invariant $AD - BD = DE$

Apply the sine rule in triangle ABD and in triangle AED , using (4):

$$\frac{AD}{\sin \angle ABD} = \frac{BD}{\sin \angle BAD}, \quad \frac{AD}{\sin \angle AED} = \frac{DE}{\sin \angle DAE},$$

that is,

$$\frac{AD}{\sin(120^\circ - \alpha)} = \frac{BD}{\sin(60^\circ - \alpha)} = \frac{DE}{\sin \alpha}. \quad (5)$$

Hence

$$AD = DE \cdot \frac{\sin(120^\circ - \alpha)}{\sin \alpha}, \quad BD = DE \cdot \frac{\sin(60^\circ - \alpha)}{\sin \alpha},$$

and, by the sum-to-product identity $\sin U - \sin V = 2 \cos \frac{U+V}{2} \sin \frac{U-V}{2}$ with $U = 120^\circ - \alpha$, $V = 60^\circ - \alpha$,

$$\sin(120^\circ - \alpha) - \sin(60^\circ - \alpha) = 2 \cos(90^\circ - \alpha) \sin 30^\circ = \sin \alpha.$$

Therefore

$$\boxed{AD - BD = DE \cdot \frac{\sin \alpha}{\sin \alpha} = DE.} \quad (6)$$

The parameter α has cancelled completely: (6) is a relation among the three lengths alone, valid for every admissible configuration. With the given data,

$$AD = BD + DE = 8 + 7 = 15, \quad DC = AD = 15, \quad BC = 23. \quad (7)$$

(For the record, α is in fact pinned down by the data: (5) gives $7 \sin(60^\circ - \alpha) = 8 \sin \alpha$, i.e. $\tan \alpha = \frac{7 \sin 60^\circ}{8 + 7 \cos 60^\circ}$, so $\alpha = 27.7958^\circ$. It is simply never needed.)

3. The hidden circle and the point P

Let

$$\Gamma = \text{circle centred } D \text{ of radius } AD = 15.$$

By (7) both A and C lie on Γ . Let P be the second intersection of line AB with Γ .

Since P, A, C all lie on Γ and B lies on segment AP , the inscribed angle at A subtending the arc PC is

$$\angle PAC = \angle BAC = 60^\circ,$$

so the corresponding central angle is

$$\angle PDC = 2 \cdot 60^\circ = 120^\circ. \quad (8)$$

Because B, D, C are collinear, the rays DB and DC are opposite, whence

$$\angle PDB = 180^\circ - 120^\circ = 60^\circ. \quad (9)$$

Now the law of cosines in triangle PDB , with $PD = 15$, $BD = 8$ and the included angle (9):

$$PB^2 = PD^2 + BD^2 - 2 \cdot PD \cdot BD \cos 60^\circ = 225 + 64 - 15 \cdot 8 = 169,$$

so

$$\boxed{PB = 13.} \quad (10)$$

The appearance of a Pythagorean-looking integer is not an accident; see Section 5.

4. The power of the point B

The point B lies inside Γ , since $BD = 8 < 15$. Its power with respect to Γ has absolute value

$$|\text{pow}_\Gamma(B)| = AD^2 - BD^2 = (AD - BD)(AD + BD) = 7 \cdot 23 = 161. \quad (11)$$

Since the line through B meets Γ at A and P , the intersecting-chords theorem gives

$$AB \cdot PB = 161.$$

With (10),

$$AB = \frac{161}{13} = 12.384615 \dots$$

Note that (11) is expressed entirely through the invariant (6): the two factors are $AD - BD = DE = 7$ and $AD + BD = DE + 2BD = 23$. So the final answer is a function of BD and DE alone,

$$\boxed{AB = \frac{DE(DE + 2BD)}{PB}, \quad PB = \sqrt{(BD + DE)^2 - BD(BD + DE) + BD^2}.$$

5. Solution B: the Eisenstein norm

Step (10) can be recognised rather than computed. Let $\omega = e^{2\pi i/3}$ and let

$$N(a + b\omega) = (a + b\omega)\overline{(a + b\omega)} = a^2 - ab + b^2$$

be the norm on the ring of Eisenstein integers $\mathbb{Z}[\omega]$. Geometrically, N is exactly the squared third side of a triangle with sides a, b and included angle 60° , which is what the law of cosines computes in (10). Hence

$$PB^2 = N(15 + 8\omega) = 15^2 - 15 \cdot 8 + 8^2 = 225 - 120 + 64 = 169 = 13^2.$$

This explains the integrality. The norm form $a^2 - ab + b^2$ is multiplicative, and its values are exactly the Loeschian numbers; that $N(15 + 8\omega)$ is a perfect square is the statement that $(15, 8, 13)$ is an “Eisenstein triple”, the 60° analogue of a Pythagorean triple. The problem was clearly designed around this triple: the data $BD = 8$, $DE = 7$ were chosen so that $AD = 15$ and $15^2 - 15 \cdot 8 + 8^2$ is a square.

(a, b)	$N(a + b\omega) = a^2 - ab + b^2$	60° -triple?
$(15, 8)$	$169 = 13^2$	yes, the one used here
$(8, 3)$	$49 = 7^2$	yes
$(5, 3)$	19	no
$(7, 5)$	39	no
$(16, 5)$	201	no
$(35, 11)$	$961 = 31^2$	yes

6. What the solution used and what it did not

The two 60° angles play different roles. The angle $\angle ADE = 60^\circ$ is used only in Section 2, to produce $\angle AED = 120^\circ - \alpha$ and hence the coincidence (4). The angle $\angle BAC = 60^\circ$ is used only in Section 3, to convert an inscribed angle into the central angle 120° . Neither is used twice.

The isosceles condition is what creates the circle. $AD = DC$ does two jobs: it gives the exterior-angle relation (1), and it places C on the circle Γ centred at D through A . Without it there is no natural circle in the picture, and the power-of-a-point step would have nothing to act on.

α is never computed. Both halves of the solution are α -free: (6) because of the sum-to-product cancellation, and Sections 3–4 because they involve only lengths and the angle $\angle PDB = 60^\circ$. This is the structural reason the answer is a clean rational number even though the configuration itself involves $\alpha = 27.7958^\circ$, an angle with no closed form worth writing.

Consistency check. Placing $D = (0, 0)$, $B = (-8, 0)$, $C = (15, 0)$ and $A = 15(\cos(180^\circ - 2\alpha), \sin(180^\circ - 2\alpha)) = (-8.47633, 12.37545)$ with $\alpha = 27.79577^\circ$, one computes

$$\angle BAC = 60.0000^\circ, \quad E = (3.02367, 6.31327) \in AC, \quad DE = 7.0000, \quad \angle ADE = 60.0000^\circ,$$

$$P = (-7.5, -12.99038), \quad PD = 15, \quad PB = 13, \quad AB = 12.3846154 = \frac{161}{13},$$

confirming every step numerically.